

BFS

```
0.47
@20525pcs003-bit →/workspaces/AI-ML (main) $ python BFS
BFS Traversal:
FireStation BuildingA BuildingB Floor1 Floor2 Floor3 @20525pcs003-bit →/workspaces/AI-ML (main) $
```

DFS

```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python DFS
Following is the Depth First Search Traversal:
['Router', 'PC1', 'Printer', 'PC2', 'Server']
@20525pcs003-bit →/workspaces/AI-ML (main) $
```

A*

```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python A*
Random Rabbits:
[Rabbit with 42 carrots., Rabbit with 9 carrots., Rabbit with 17 carrots., Rabbit with 46 carrots., Rabbit with 36 carrots.]

Carrots of First Rabbit: 42

Best Rabbit: Rabbit with 46 carrots.
@20525pcs003-bit →/workspaces/AI-ML (main) $
```

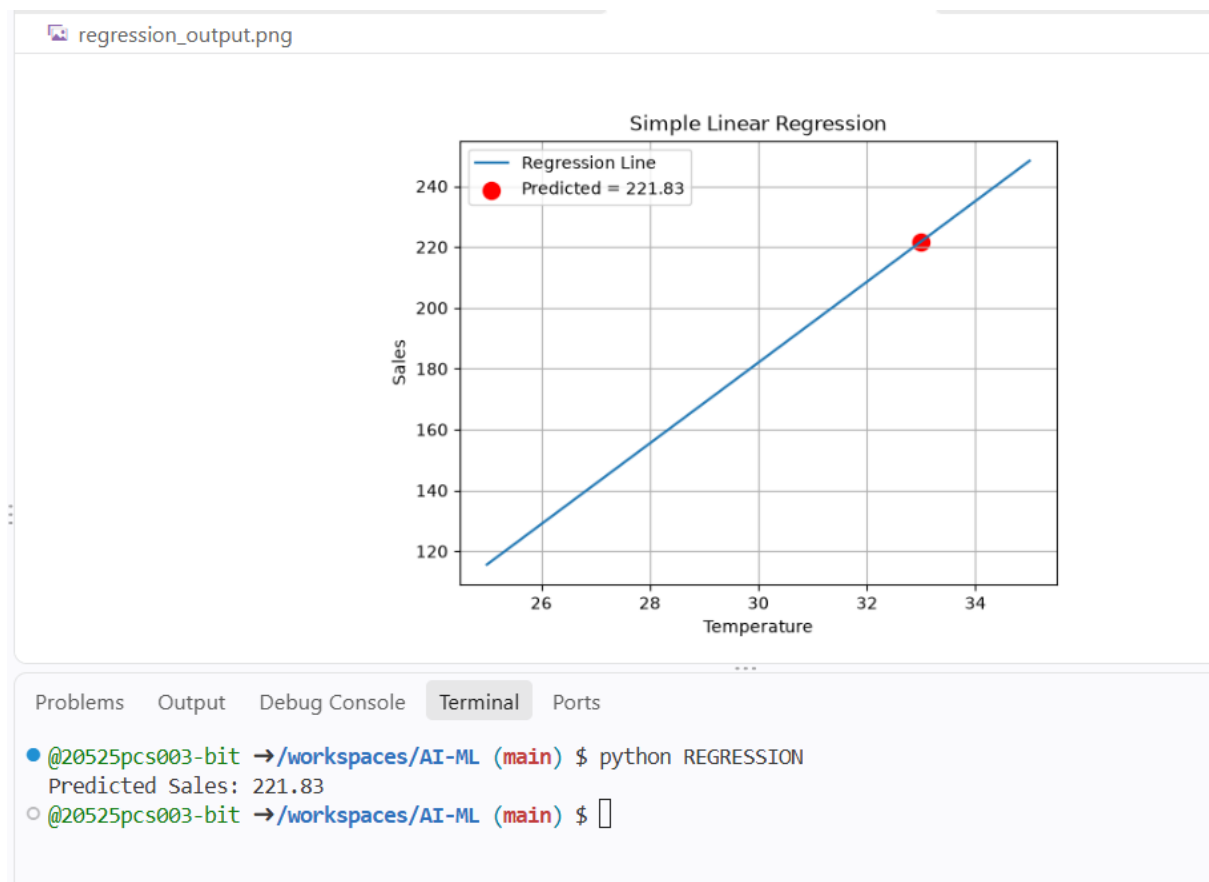
NAÏVE BAYES

```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python NAIVEBAYES
Has good programming skills? (yes/no): yes
Selected = 0.82
Rejected = 0.18
Prediction: SELECTED
@20525pcs003-bit →/workspaces/AI-ML (main) $
```

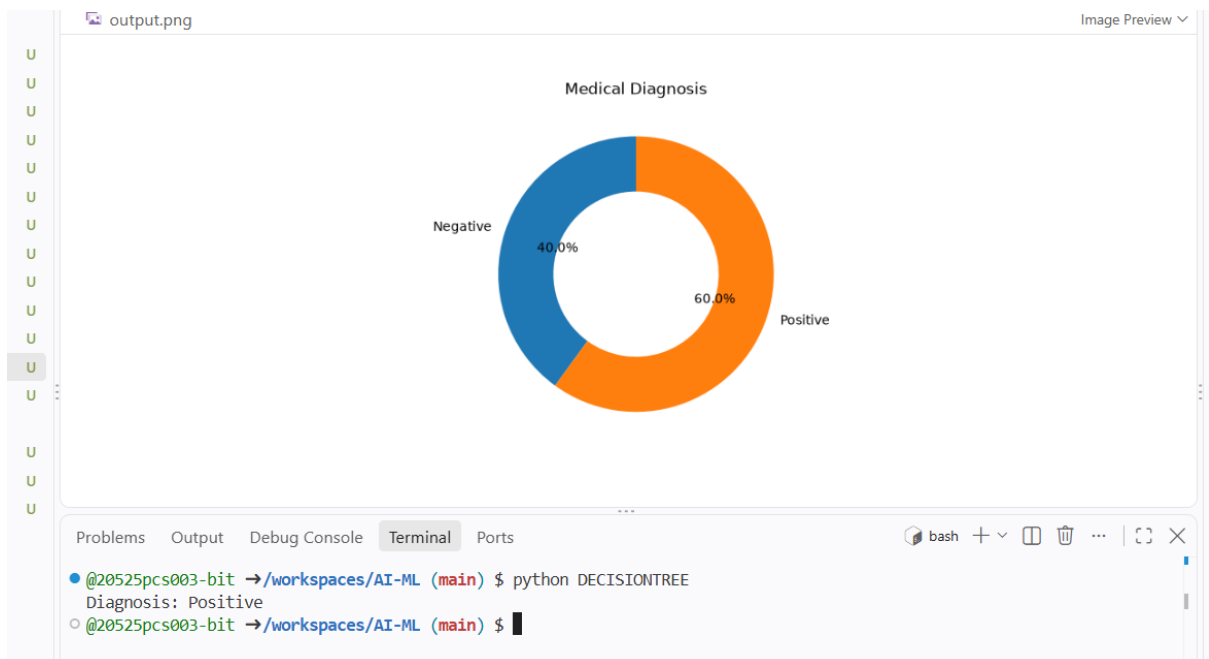
BAYESIAN NETWORK

```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python BAYESIAN
@20525pcs003-bit →/workspaces/AI-ML (main) $ python BAYESIAN
CCTV Security
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P(Intruder) = 0.1
P(Motion|Intruder) = 0.95
P(Motion) = 0.2
Intruder Probability:
0.47
@20525pcs003-bit →/workspaces/AI-ML (main) $
```

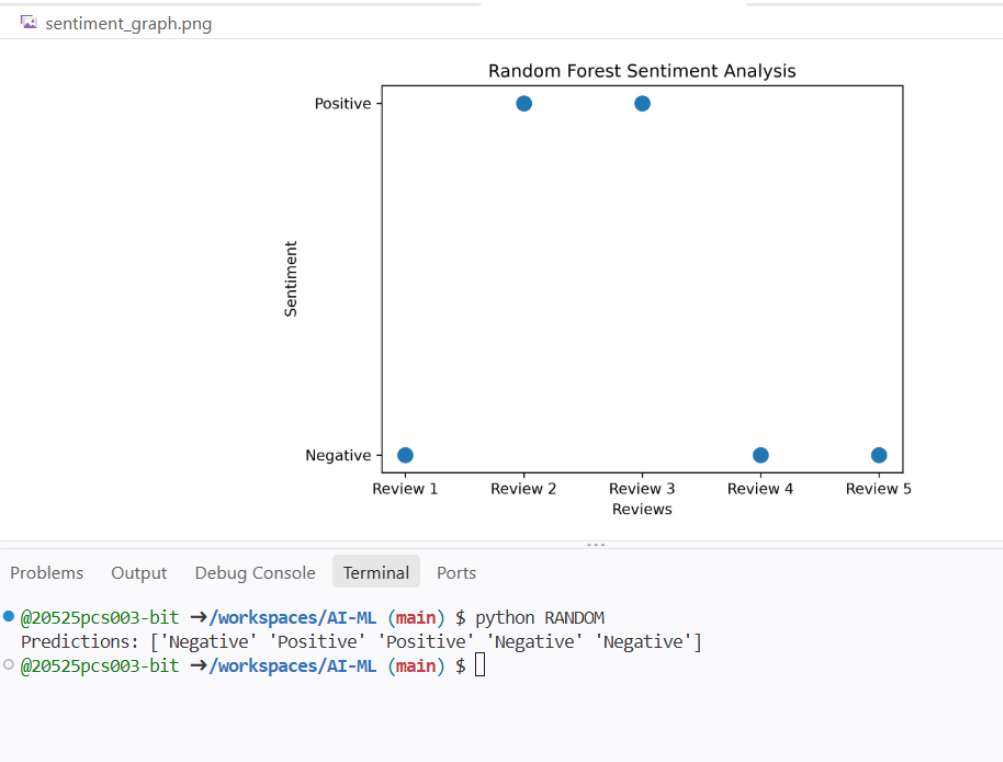
REGRESSION MODEL



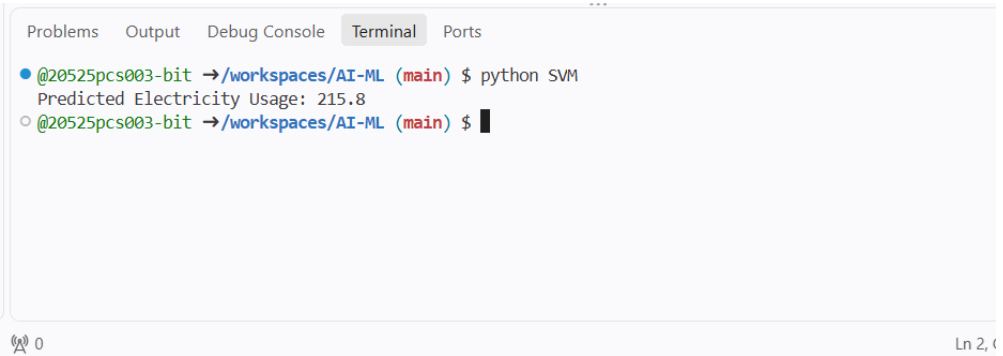
DECISION TREE



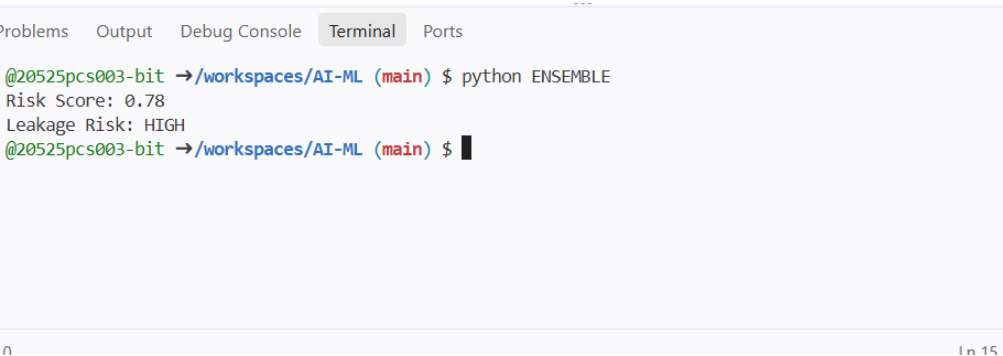
RANDOM FOREST



SVM MODELS



ENSEMBLE TECHNIQUES



CLUSTERING ALGORITHM

```
Problems  Output  Terminal  Ports

● @20525pcs003-bit →/workspaces/AI-ML (main) $ python CLUSTERING
User 1 -> Group 3
User 2 -> Group 3
User 3 -> Group 3
User 4 -> Group 2
User 5 -> Group 2
User 6 -> Group 2
User 7 -> Group 1
User 8 -> Group 1
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```

EM BAYESIAN NETWORK

```
Problems  Output  Debug Console  Terminal  Ports

● @20525pcs003-bit →/workspaces/AI-ML (main) $ python EMBAYESIAN
Temperature: 44 °C
Usage: 8 hours
Power: 2.4 kW
Failure Risk: 1.0
Result: HIGH RISK
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```

NN MODELS

```
Problems  Output  Debug Console  Terminal  Ports

● @20525pcs003-bit →/workspaces/AI-ML (main) $ python NNMODELS
Congestion Predictions: [1 1 1 1 1]
Congestion Probability: 1.0
Prediction: HIGH CONGESTION
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```

DEEP LEARNING NN MODELS

```
Problems  Output  Debug Console  Terminal  Ports

● @20525pcs003-bit →/workspaces/AI-ML (main) $ python DEEPLARNINGNNMODEL
Failure Probability: 1.0
Prediction: FAILURE
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```